

GapMind prediction (*in silico*)

Pathway predicted present (+)

Pathway predicted absent (–)

Experimental outcome (*in vivo*)

Growth observed (+)
No growth (–)

a TRUE POSITIVE

concordant

BIOLOGICAL STATE

Genes present and functional
Pathway correctly annotated as present
Experiment accurately measures growth

INTERPRETATION

Prediction and phenotype agree:
gene-to-function mapping is supported.

High-quality training sample

b FALSE NEGATIVE

discordant

POSSIBLE CAUSES

- ANNOTATION homolog not detected
- BIOLOGY alternative pathway present
- BIOLOGY promiscuous enzyme activity
- MEASUREMENT false-positive growth call
- MEDIA growth on base media (carryover)

Growth via unknown mechanism

c FALSE POSITIVE

discordant

POSSIBLE CAUSES

- REGULATION pathway repressed under condition
- MEDIA insufficient incubation time
- MEDIA missing cofactors / inducers
- ANNOTATION false-positive homology call
- BIOLOGY incomplete pathway (missing step)

Genes present but not functional

d TRUE NEGATIVE

concordant

BIOLOGICAL STATE

Genes absent from genome
Pathway correctly annotated as absent
No alternative pathway or leaky activity
Experiment accurately measures no-growth

INTERPRETATION

Absence of capability confirmed by both
genome content and phenotype.

High-quality training sample